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Experimentally based evaluation of accuracy of absorption equilibrium measurements

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Abstract

The accuracy of absorption equilibrium measurements for the system CO₂-water-Monoethanolamine (MEA) has been discussed by various authors in the context of choosing data to fit parameters to proposed mathematical / thermodynamic models to represent the measured data. The reported data for this system have been generated from 1935. Since absorption equilibrium values directly influence the mass transfer driving force and thus the height of an absorption column, it is of great interest to establish a better base for judging the uncertainty of absorption equilibrium data. In this work a set of absorption experiments were run with 30%(wt) MEA to perform equilibrium measurements for CO₂-water-MEA system at atmospheric pressure, 40°C temperature and different CO₂ loading conditions. The results were compared with the literature values, and a detailed uncertainty analysis was done for the final results by using GUM and QUAM methods.

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1. Introduction

Absorption of CO₂ in to amine-based solutions and stripping them by using heat is a mature technology, which is many decades old. Several studies have been carried out on the solubility of CO₂ in aqueous Monoethanolamine (MEA) solution. Mason and Dodge[1] studied and reported equilibrium absorption of CO₂ by solutions of ethanolamine already in 1936.

Uncertainty analysis was done by some researchers and different standard were followed. Choosing data to fit parameters to proposed mathematical/thermodynamic models to represent the measured data were discussed by various authors. It is noticed that the authors do not necessarily choose the same data to use or exclude. In general, parameters in models are found by regression analysis; the data are then compared with the fitted model. Data points that deviate from the model by more than a certain percentage are then discarded and a new regression analysis is performed. The requirements to report on expected measurement uncertainty was less stringent in the early days than today. Since absorption equilibrium values directly influence the mass transfer driving force and thus the height of an absorption

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column, it is of great interest to establish a better base for judging the uncertainty of absorption equilibrium data. Any uncertainty in an estimate will in a well engineered absorption column lead to extra design margins being added, which in turn leads to a taller and more expensive column.

1.1. Literature review on CO₂-water- MEA equilibrium

The data for CO₂ solubility in aqueous MEA reported by Mason and Dodge [1] and Reed and Wood [2] did not claim high accuracy for their data at that time. Similar studies were made by Reed et al. [3], Lyudkovskaya and Leibush[4] and Muhlbauer and Monaghan[5]. There were considerable difference in the reported values and the data in the low partial pressure and the high temperature region were insufficient. A detailed study regarding CO₂-H₂O-MEA equilibrium was done by Atadan[6] but proper uncertainty analysis was not reported except the statistical analysis regarding the deviation of their results from the least square straight line in logarithmic plots. Jones et al.[7] claimed to reported CO₂ solubility in aqueous MEA [15.3% (wt)] with a high accuracy at that time, but the uncertainty of their solubility data was not mentioned. They have worked in the 40°C to 140°C temperature range with an accuracy of $\pm 0.1^\circ\text{C}$ to $\pm 0.5^\circ\text{C}$ respectively, but the accuracy of pressure measurements was not reported.

Lee et al.[8]&[9] reported the equilibrium data obtained for CO₂ in 6.1%(wt),15.3%(wt),22.9%(wt) and 30.5%(wt) MEA solutions. The data were smoothed by the preparation of cross-plots with temperature and aqueous MEA concentration. Agreement between their data and already published data by that time was good. The major errors in their work were associated with the measurement of the equilibrium pressure and the determination of the pressure and volume of the gas evolved from the acidified liquid samples. Lawson et al.[10] studied the solubility of CO₂, H₂S and their mixtures in aqueous MEA and DEA solutions. Accuracy of the data was estimated around 10% and analysis procedure was not given in detail. There was a detailed study done by Jou et al.[11] in a wide temperature range from 0°C to 150°C with 0.001kPa to 19.9kPa partial pressure. Many research groups have used those data until now for both research and modelling purposes. A gas chromatograph was used to analyse both the liquid and the gas samples, but in some cases liquid samples were analysed by the BaCO₃ method. Even though their analysis showed that the error in the BaCO₃ method is $\pm 3\%$, and that in the gas chromatograph method is $\pm 2\%$, details of their error analysis were not stated. Isaacs et al.[12] reported the equilibrium data and the data accuracy from a 15.2%(wt) aqueous MEA-CO₂ system. Total accuracy of their results is given as $\pm 15\%$ but the analysis procedure was not reported. Our literature review has shown that a detailed total uncertainty analysis for the experimentally obtained in the gas phase and the in the liquid phase at the equilibrium conditions were not reported for the aqueous MEA-CO₂ equilibrium studies from 1936 to 2011 except Ma'mum et al.[13]. Total uncertainty of the equilibrium data were given by Ma'mum et al.[13] as $\pm 2\%$ for CO₂ loading measurements based on relative standard uncertainty of two to five parallel samples. The estimated relative expanded uncertainty in the CO₂ partial pressures was also found as $\pm 2\%$.

Most of the authors who worked with the aqueous MEA-CO₂ equilibrium studies have used the reported equilibrium data from previous authors to estimate the accuracy of their results such as Lawson et al.[10], Isaacs et al.[12], Shen and Li.[14], Jou et al.[11], Alaei et al.[15], Ma'mum et al.[13], and Xu and Rochelle[16]. It is possible to observe that authors Shen and Li[14], Park et al.[17], Portugal et al.[18] and Aronu et al.[19] have reported the individual uncertainty of their temperature, pressure and weight measurements in detail based on their experimental procedure but not as a total uncertainty of the in the gas phase and the in the liquid phase at the equilibrium conditions. A number of authors like Park et al.[17], Portugal et al.[18], Xu and Rochelle[16] and Aronu et al.[19] were more focused on validating their experimental results based on mathematical models but attention to a total uncertainty analysis was missing. A detailed total uncertainty analysis for the experimentally obtained in the gas phase and the in the liquid phase at the equilibrium conditions were reported by Tong et al.[20].

They studied the aqueous MEA-CO₂ equilibrium by using 30 % (wt) MEA at 40°C and 120°C temperature within the CO₂ partial pressure range of 3.95-295.2 kPa, and 0.075-0.540 CO₂ loadings.

2. Equilibrium experiment

The gases (N₂ and CO₂) were supplied from the laboratory gas system through the gas flow controllers (Sierra). Purities of the N₂ and CO₂ is approximately 99.99%. N₂ and CO₂ were supplied by Yara Praxair AS. Several gas concentrations were used to calibrate the GC and they were supplied by AGA Gas. According to the supplier, (Merck KGaA, Germany) the purity of the MEA is better than 99.5%. All the gravimetric operations were carried out using a Mettler XS-403S precision balance from Mettler Toledo with ± 0.001 g accuracy. Trace GC×GC gas chromatography system from Thermo-Scientific (model: KAV00349) was used to analyse the CO₂ concentration in gas samples. Liquid sample analysis was done by using a Mettler Toledo T50 titrator with DG300SC and DX200 electrodes. Hardi automobile electronic fuel pump (model: 18812) was used to circulate the gas mixture through the system with flow rate.

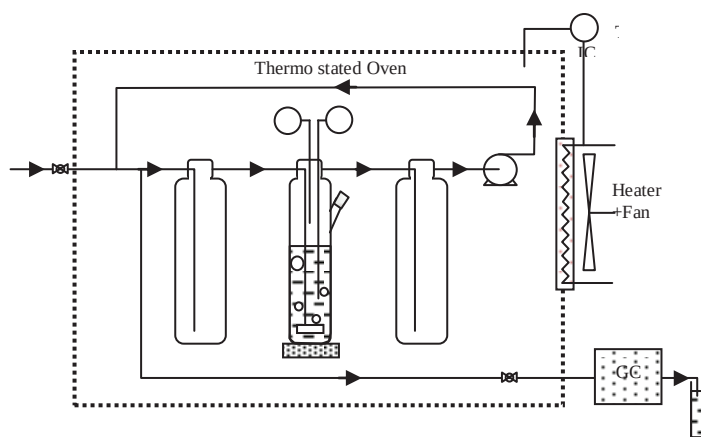


Fig.1. Vapour-liquid equilibrium apparatus

At the equilibrium state, the concentration of CO₂ was measured for both in the liquid and gas phases at the atmospheric pressure and 40°C temperature, and the concentrations of aqueous MEA was 30% (wt).

The vapour-liquid equilibrium apparatus (See fig.1) was built by studying the low temperature /atmospheric vapour-liquid apparatus used in Ma'mum et al.[13]&[21] and Aronu et al.[19]. Ma'mum et al.[21] explained about the apparatus, It was contained in a water bath and equilibrium cells were partially submerged in the water bath to maintain a constant temperature. Here in this setup the water bath was replaced with a high-quality temperature controller (West 6100 from West instruments-Process controls) with a PT100 thermal element to control an air bath. The IR CO₂ analyser was replaced with a gas chromatograph (GC). Even though the apparatus consists of 3 equilibrium cells, in this work only one of them (number 2, See fig.1) was containing the aqueous MEA solution in a temperature controlled environment. The others were used for scrubbing purposes.

The experiments were performed at nearly atmospheric pressure. The closed loop pressures were measured by an Endress + Hauser, Cerabar S pressure transmitter and the atmospheric pressure was measured by using a Weems & Plath, Electronic barometer. Aqueous MEA solution (180) was prepared by a gravimetric method and transferred to the equilibrium cell. Same set up was used to load the aqueous MEA solution with CO₂.

The system has to be flushed with N₂ to remove the dissolved oxygen in loaded MEA solution. After 5 minutes the outlet was closed and the system was pressurised to with a 95% N₂ and 5% CO₂ gas mixture. The apparatus temperature set point was fixed to 40°C and gas phase was circulated typically for 2h after the temperature has reached to the set point. The CO₂ concentration in the gas phase was measured by the GC through 4 gas samples 20 ×4) taken from the closed loop. Several certified calibrations gases were used to calibrate the GC depending on the CO₂ concentration of the samples as shown in table 1. The total concentration range was divided into 4 difference ranges and separate calibration curves were created for each range by using different calibration gases.

Soon after partial pressure was measured, a liquid sample (50ml) was taken from the equilibrium cell. The liquid phase CO₂ loading analysis was done by using the BaCl₂ method.

Table 1: Usage of the calibration gases

Range of the CO ₂ concentration (%)	Number of calibration points being used to create the calibration curve				
	(0,0)	0.085% ± 2% relative	0.98% ± 2% relative	7.09% ± 2% relative	15% ± 2% relative
0.02-0.085	1	3	-	-	-
0.085-0.98	-	3	3	-	-
0.98-7	-	-	3	3	-
7-20	-	-	-	3	3

3. Measurements uncertainty

The formal procedure for estimating uncertainty is relatively new in the history of data measurement. It is determined that even if an appropriate correction have been made for suspected components of error, there still remains an uncertainty about the accuracy of the stated measurements. Therefore it is important to present some quantitative indication of the quality of the results when publishing the result of a measurement of a physical quantity. It will help to compare the results of own work with those of others.

The Bureau International des Poids et Mesures (BIPM) published a guide book in 1995 (published in 1993, corrected and republished in 1995)[22] together with another seven supporting organizations including International Organization of Standardization (ISO). The guide has republished again in 2008(JCGM 100:2008)[23] with some minor corrections. The guide establishes general rules for evaluating and expressing uncertainty in measurement that are intended to be applicable to a broad spectrum of measurements[22]. This is known as GUM for short.

Uncertainty of the measurement defined as, a ‘*parameter which associated with the result of a measurement, that characterizes the dispersion of the value that could reasonably be attributed to the measurand*’ [22].

The guide provides several types of uncertainty calculations based on the calculation method and the nature of the measurements. The only drawback with GUM was, that it was not that applicable for calculating the uncertainty of analytical measurements. To fulfil that requirement a document called ‘‘Quantifying Uncertainty in Analytical Measurement’’ (QUAM) published in 1995[24]. The second edition of the QUAM guide was prepared with the practical experience of uncertainty estimation in chemistry laboratories and published in 2000. This guide is explicitly targeted for uncertainty measurements of chemical analysis in full compliance with formal ISO guide principles [24].

In this work all the measurements are summarized into experimentally obtained in the gas phase and the in the liquid phase at the equilibrium conditions.

With the nature of the and measurements GUM and QUAM methods were used for uncertainty calculations for and respectively.

3.1. Uncertainty analysis for experimentally obtained in the gas phase

The standard uncertainty in the CO₂ partial pressure measurements was calculated based on Eq. (1), taking into account the contribution of the uncertainty of the temperature measurement, MEA concentration measurement due to errors by weighing the pure MEA to prepare the aqueous MEA solution, total pressure measurement, and the uncertainty involved in the calibration of the gas-chromatography. Hence,

[T- Temperature at the equilibrium conditions, – Concentration of the aqueous MEA solution, P- Total pressure in the system, A- Peak area (GC)] By applying the Equation 11a from JCGM 100:2008 [23],

$$\text{---} \quad (1)$$

, where is the combined standard uncertainty of the partial pressure of the CO₂ in the gas phase. The experiments are run almost at the atmospheric pressure and the uncertainty based on the total pressure changes in the system was neglected.

Therefore,

$$\text{---} \quad \text{and}$$

Measurements uncertainty was calculated for the worst case scenario. The uncertainty of the MEA concentration is mainly due to the uncertainty of the weight measurements of the MEA which was used to make the 30%(wt) MEA solution. The is calculated to be below 0.0008mass% and its effect for the standard uncertainty in the CO₂ partial pressure measurements is negligible.

Uncertainty of the temperature measurement is and is the standard uncertainty for the calibration of the gas chromatographs. is calculated based on the method explains in QUAM, which is calculated to be below 3% from measured partial pressure. The standard uncertainty in the CO₂ partial pressure measurements is calculated for each equilibrium measurements for 30%(wt) MEA at 40°C and shown in chapter 4.

3.2. Uncertainty analysis for experimentally obtained CO₂ loading measurements.

The QUAM method was used for calculating the combined standard uncertainty in the CO₂ in the liquid phase. This method describes a four step procedure for calculation of measurement uncertainty. All the evaluated sources and their influence of the parameters are shown in the cause and effect diagram, Fig.2.

The BaCl₂- method used for analysing the CO₂ loading in the liquid phase consists of four main steps that may contribute to uncertainty. The main steps are; weighting of the sample (m_{sample}), heating and filtration, titration to pH=2 and titration to the equivalence point. For the two titration steps both volume (V_{HCl} and V_{NaOH}) and concentration (C_{HCl} and C_{NaOH}) of the titrant solution have influence in the uncertainty. The uncertainty contribution in the molar mass of MEA, M_{MEA} is neglectable in comparison

to the other sources. In addition uncertainty in concentration of MEA, C_{MEA} was calculated separately for all the sources shown in the branch to the right in the cause and effect diagram, using the QUAM method. This was found to be 0.23%. The repeatability factor of the method includes the combined repeatability contribution of all the parameters. This is found by several repeated measurements performed by three persons.

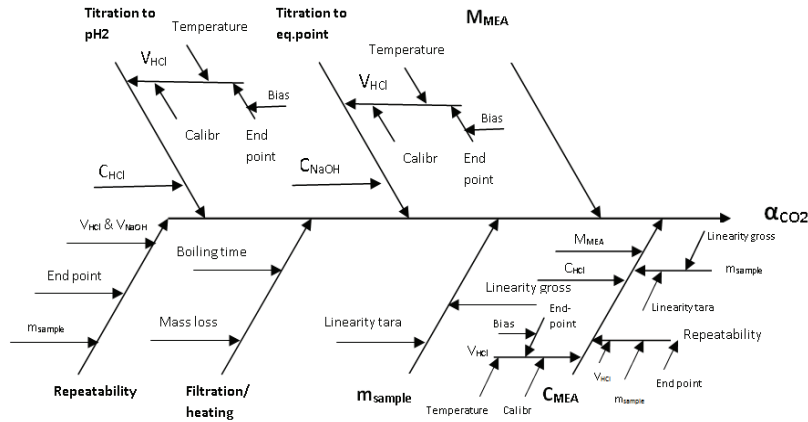


Fig.2. Cause and effect diagram with the identified uncertainty sources

The combined standard uncertainty in the loading analysis was calculated using Equation 2, where the relevant uncertainty sources from the cause and effect diagram are taken into account [24].

$$u_c(\alpha_{CO_2}) = \alpha_{CO_2} \cdot \sqrt{\sum_{i=1}^n \left(\frac{u(x_i)}{x_i} \right)^2}$$

$$= \alpha_{CO_2} \cdot \sqrt{rep^2 + \left(\frac{u(m_{sample})}{m_{sample}} \right)^2 + \left(\frac{u(C_{MEA})}{C_{MEA}} \right)^2 + 2 \cdot \left(\frac{u(V_{HCl})}{V_{HCl}} \right)^2 + 2 \cdot \left(\frac{u(C_{HCl})}{C_{HCl}} \right)^2 + 2 \cdot \left(\frac{u(V_{NaOH})}{V_{NaOH}} \right)^2 + 2 \cdot \left(\frac{u(C_{NaOH})}{C_{NaOH}} \right)^2} \quad (2)$$

Where $\frac{u(x_i)}{x_i}$ are the relative uncertainties in the contributing parameters from the cause and effect diagram, expressed as relative standard deviation. α_{CO_2} is the CO₂ loading and $u_c(\alpha_{CO_2})$ is the combined standard uncertainty in the CO₂ loading. rep is the combined repeatability of the parameters, m_{sample} is the sample mass, V_{HCl} and V_{NaOH} is titration volume of HCl and NaOH respectively. C_{HCl} and C_{NaOH} is the concentration of HCl and NaOH. The uncertainty values for the volume and concentration of HCl and NaOH is multiplied by two, because of the contribution of uncertainty for both sample and blank sample.

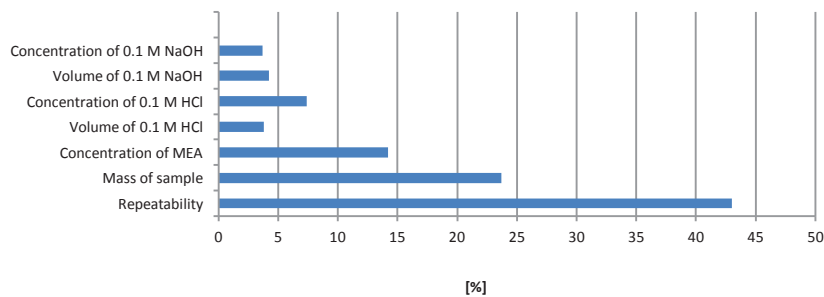


Fig. 3. Contributions of uncertainty for the different parameters in the CO₂ loading analysis

The combined uncertainty for the CO₂ loading analysis was found to be 1.3 %. The contributions of the different parameters are shown in Fig.3.

4. Results and discussion

Experimentally measured equilibrium data for 30%(wt) aqueous MEA at 40°C with their uncertainties are shown in the table 2. All the data are plotted with the literature values in the figure 4 with the upper and lower limit of the results from this work based on the uncertainty calculations. Literature values are shown in the table 3. These are the only literature data found for water-MEA-CO₂ equilibrium system for 30%(wt) MEA at 40°C. Figure 4 is zoomed out to the loading region 0.2-0.6 and shown in the figure 5 for better resolution.

Table 2 .Vapour liquid equilibrium data from this work for 30 %(wt) aqueous MEA at 40oC

CO ₂ partial pressure (bar)	CO ₂ loading (mol/kg MEA)	CO ₂ loading (mol/kg MEA)	CO ₂ loading (mol/kg MEA)	CO ₂ loading (mol/kg MEA)	CO ₂ loading (mol/kg MEA)
0.0099	0.197	0.0083	0.0115	0.194	0.200
0.0146	0.213	0.0129	0.0162	0.210	0.216
0.0327	0.300	0.0307	0.0343	0.296	0.304
0.0515	0.332	0.0491	0.0531	0.328	0.336
0.2433	0.405	0.2346	0.2449	0.400	0.410
0.4351	0.437	0.4198	0.4367	0.431	0.443
2.9370	0.500	2.8342	2.9386	0.494	0.506
10.269	0.527	9.9096	10.271	0.520	0.534
15.593	0.540	15.047	15.595	0.533	0.547

are the maximum possible error of the measurements which are calculated based of the uncertainty analysis for experimentally obtained CO₂ partial pressure in the gas phase and the CO₂ loading in the liquid phase, respectively.

Table 3.Vapour liquid equilibrium data from literature for 30 %(wt) aqueous MEA at 40°C.

Lee et al.(1974)[8]		Lee et al.(1976)[9]		Shen et al.(1992)[14]		Jou et al.(1995)[11]		Aronu et al.(2011)[19]		Tong et al. (2012)[20]	
5973.214	0.966	0.1	0.325	2.2	0.471	19914	1.18	0.0016	0.102	3.95	0.53
1906.521	0.829	0.316	0.379	5	0.496	14945	1.132	0.0123	0.206	19.12	0.585
776.118	0.724	1	0.427	12.8	0.512	9969	1.097	0.0246	0.25	71.5	0.632
284.669	0.676	3.16	0.468	28.7	0.538	5986	1.049	0.0603	0.337	101	0.639
135.786	0.625	10	0.508	58.4	0.57	2992	0.965	0.0851	0.353	159.4	0.668
82.712	0.582	31.6	0.55	101.3	0.594	993	0.844	0.1835	0.401	161.52	0.67
10.753	0.491	100	0.603	140.1	0.62	593	0.794	0.2928	0.417	211.92	0.687
2.805	0.46	316	0.681	552	0.676	293	0.709	0.3188	0.421	297.02	0.706
		1000	0.78	883	0.728	103	0.646	0.3809	0.433	408.17	0.748
		3160	0.898	1256	0.763	36.1	0.609	0.5702	0.447	3.95	0.53
		10000	1.05	1580	0.772	8.09	0.557	1.0662	0.464	19.12	0.585
		0.1	0.325	1973	0.806	2.57	0.513	1.8326	0.476	71.5	0.632
						0.604	0.461	1.8278	0.477	101	0.639
						0.068	0.365	2.3193	0.485	159.4	0.668
						0.009	0.203	2.8577	0.489	161.52	0.67
						0.001	0.0888	8.5583	0.516		
								11.812	0.524		

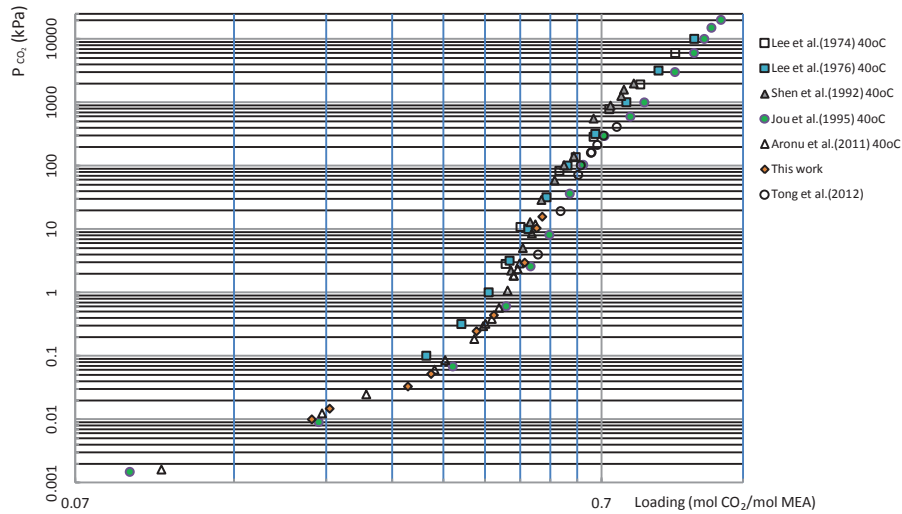


Fig.4. Equilibrium solubility, 30%(wt) aqueous MEA at 40°C

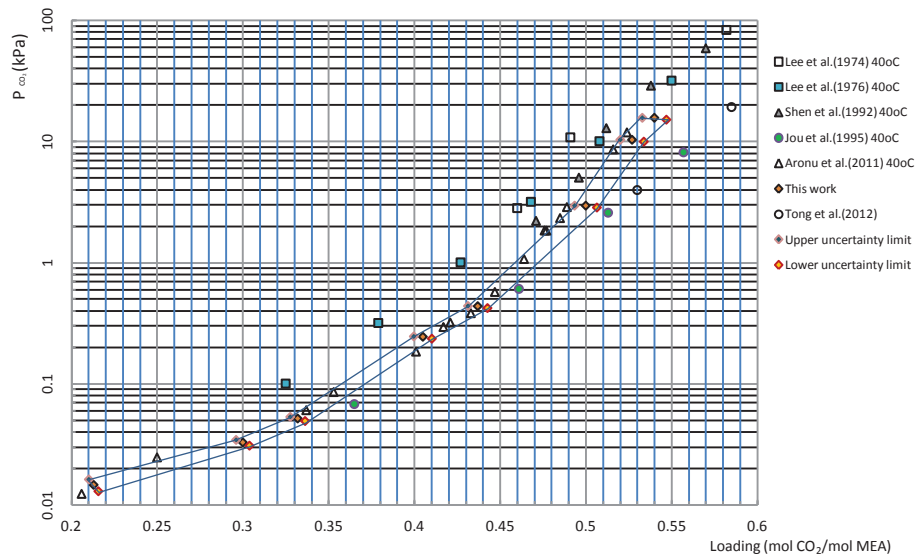


Fig.5. Equilibrium solubility, 30%(wt) MEA aqueous at 40°C in region of 0.2-0.6

loading.

As seen from figure 5, the experimental results from this work agreed well with the lately published water-MEA- CO₂ equilibrium data from Aronu et al.[19]. The equilibrium data from Shen and Li[14] also having a good agreement, unfortunately their research was more focused on higher loading condition than 0.47

It is possible to observe that the early published data from Lee et al[8]&[9] deviated more from the results from this work than the other literature values. Jou et al.[11] claimed that

the data from Lee et al.[8]&[9] was neglected the remaining CO₂ in the acidic solution in their analysis of the liquid phase loading.

Equilibrium data published by Jou et al.[11] and Tong et al.[20]. claimed high accuracy. Still those data shows continuous offset compared to the results from the present work and does not even lay in the region of lower and upper uncertainty limit of these results. Most probably Shen and Li[14], Jou et al.[11], Tong et al.[20], and Aronu et al.[19] had a similar measurements uncertainty for their data as the uncertainty of the results from this work.

There is a good agreement between the data from Jou et al.[11] and Tong et al.[20]. Jou et al.[11] used the gas chromatograph to measure the CO₂ partial pressure and liquid phase analysis was also done by a chromatographic technique in some cases in addition to the BaCl₂ method. Tong et al.[20] have used the gas chromatograph for both partial pressure and loading measurements.

Equilibrium data published by Aronu et al.[19] have used the same measurements technique and very similar experiment setup as in this work and also it shows a very good agreement in between.

V.R. Meyer[25] mentioned that different standard deviations in measurements could be observed for the same sample, if the sample is analyse in different days or different laboratories. This means there are more reasons for difference between reported vapour liquid equilibrium data for H₂O-CO₂-MEA system, than the calculated uncertainty. This explains the little off set between the data from present work with the other literature data from Tong et al. (2012)[20] and Jou et al.(1995)[11].

5. Conclusion

New experimental data for vapor-liquid equilibrium of CO₂ in 30%(wt) MEA solution at 40°C temperature are presented and compared with the literature values. A detailed uncertainty analysis was done for the measurements from both CO₂ partial pressures in the gas phase and CO₂ loading in the liquid phase. Generally the vapor liquid equilibrium data from this work give a good agreement with the literature values, still there are slight deviations. The explanation for this could not be merely the errors of the measurements. It is possible to conclude that the experimental method and procedure can be affected on the vapour liquid equilibrium of CO₂-water-MEA system for some extent.

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References

- [1] Mason JW, Dodge BF. Equilibrium absorption of carbon dioxide by solutions of the ethanolamines. *Trans Am Inst Chem Eng* 1936;**32**:27–48.
- [2] Reed RM, Wood W R. *Trans Am Inst Chem Eng* 1941;**37**:363-82.
- [3] Reed RM. (to Girdls Corp.). U. S. Patent 2,399,142, 1942.
- [4] Lyudkovskaya MA, Leibush AG. Solubility of Carbon Dioxide in Solutions of Ethanolamines Under Pressure. *Zh Prikl Khim*1949;**22**:558-567.
- [5] Muhlbauer HG, Monaghan PR. Sweetening Natural Gas with Ethanolamine Solutions. *Oil Gas J* 1957;**55**:139-145.
- [6] Atadan EM. Absorption of Carbon Dioxide by Aqueous Monoethanolamine Solutions. PhD. Thesis. University of Tennessee, Knoxville, TN. 1954.
- [7] Jones JH, Froning HR, Claytor EE. Solubility of Acidic Gases in Aqueous Monoethanolamine. *J Chem Eng Data* 1959;**4**:85-92.
- [8] Lee JI, Otto FD, Mather AE. The Solubility of H₂S and CO₂ in Aqueous Monoethanolamine Solutions. *Can J Chem Eng* 1974;**52**:803-805.

- [9] Lee JI, Otto FD, Mather AE. Equilibrium between Carbon Dioxide and Aqueous Monoethanolamine Solutions. *J Appl Chem Biotechnol* 1976;**26**:541-549.
- [10] Lawson JD, Garst AW. Gas Sweetening Data: Equilibrium Solubility of Hydrogen Sulfide and Carbon Dioxide in Aqueous Monoethanolamine and Aqueous Diethanolamine Solutions. *J Chem Eng Data* 1976;**21**:20-30.
- [11] Jou FY, Mather AE, Otto FD. The Solubility of CO₂ in a 30 Mass Percent Monoethanolamine Solution. *Can J Chem Eng* 1995;**73**:140-147.
- [12] Isaacs EE, Otto FD, Mather AE. Solubility of Mixtures of H₂S and CO₂ in a Monoethanolamine Solution at Low Partial Pressures. *J Chem Eng Data* 1980;**25**:118-120.
- [13] Ma'mun S, Nilsen R, Svendsen HF, Juliussen O. Solubility of carbon dioxide in 30mass% monoethanolamine and 50mass% methyl-diethanolamine solutions. *J Chem Eng Data* 2005;**50**:630-634.
- [14] Shen KP, Li MH. Solubility of Carbon Dioxide in Aqueous Mixtures of Monoethanolamine with Methyl-diethanolamine. *J Chem Eng Data* 1992;**37**:96-100.
- [15] Alaei J, Tajerian M. Experimental Equilibrium between Acid Gases & Ethanolamine Solutions. *Petrol Coal* 2001;**43**:80-84.
- [16] Xu Q, Rochelle G. Total pressure and CO₂ solubility at high temperature in aqueous amines. *Energy Procedia* 2011;**4**:117-124.
- [17] Park SH, Lee KB, Hyun JC, Kim SH. Correlation and Prediction of the Solubility of Carbon Dioxide in Aqueous Alkanolamine and mixed Alkanolamine solutions. *Ind Eng Chem Res* 2002;**41**:1658-1665.
- [18] Portugal AF, Sousa JM, Magalhães FD, Mendes A. Solubility of carbon dioxide in aqueous solutions of amino acid salts. *Chem Eng Sci* 2009;**64**:1993-2002.
- [19] Aronu UE, Gondal S, Hessen ET, Warberg TH, Hartono A, Hoff KH, Svendsen HF. Solubility of CO₂ in 15,30,45 and 60mass% MEA from 40 to 120°C and model representation using the extended UNIQUAC framework. *Chem Eng Sci* 2011;**66**:6393-6406.
- [20] Tong D, Trusler M, Maitland GC, Gibbins J, Fennell PS. Solubility of carbon dioxide in aqueous solution of monoethanolamine or 2-amino-2-methyl-1-propanol: Experimental measurements and modelling. *Int J Green Gas Cont* 2012;**6**:37-47.
- [21] Ma'mun S, Jakobsen JP, Svendsen HF, Juliussen O. Experimental and modelling study of the solubility of carbon dioxide in aqueous 30 mass% 2-((2-minoethyl)amino) ethanol solution. *Ind Eng Chem Res* 2006;**45**:2505-2512.
- [22] JCGM GUM, Guide to the expression of uncertainty in measurement edition, 1993, corrected and reprinted 1995, International Organisation for Standardisation, Geneva, Switzerland. 1995.
- [23] JCGM 100, Evaluation of measurement data - Guide to the expression of uncertainty in measurement. GUM 1995 with minor corrections and reprinted 2008, 1st edition Geneva, Switzerland. 2008
- [24] Ellison SLR, Ossleim MR, Williams A, (Eds.), EURACHEM/CITAC Guide, Quantifying Uncertainty in Analytical Measurement, 2nd ed. 2000.
- [25] Meyer VR. Measurement uncertainty. *J Chrom.A.* 2007.**1158**. 15-24.